

Fast simulations of multidimensional NMR spectra of proteins and peptides

Thomas Vosegaard 

Danish Center for Ultrahigh-Field NMR Spectroscopy, Interdisciplinary Nanoscience Center and Department of Chemistry, Aarhus University, DK-8000 Aarhus, Denmark

Correspondence

Thomas Vosegaard, Danish Center for Ultrahigh-Field NMR Spectroscopy, Interdisciplinary Nanoscience Center and Department of Chemistry, Gustav Wieds vej 14, DK-8000 Aarhus, Denmark.
Email: tv@chem.au.dk

Funding information

H2020 Future and Emerging Technologies (731475); H2020 Research Infrastructures (731005); Danish Ministry of Higher Education and Science, Grant/Award Number: AU-2010-612-181

To simulate full multidimensional nuclear magnetic resonance spectra of peptides and proteins in a reasonable time frame, a strategy for separating the time-consuming full-density matrix calculations from the chemical shift prediction and calculation of coupling patterns is presented. The simulation setup uses SIMulation Program for SOLid-state NMR (SIMPSON) to calculate total correlation spectroscopy transfer amplitudes and average distances as a source for nuclear Overhauser effect spectroscopy transfer amplitudes. Simulated ^1H 1D, 2D total correlation spectroscopy, and 2D nuclear Overhauser effect spectroscopy nuclear magnetic resonance spectra of peptides with sequence Pro-Ala-Gly-Tyr-Asn and Asn-Phe-Gly-Ala-Ile-Leu and of ubiquitin are presented. In all cases, the simulations lasted from a few seconds to tens of seconds on a normal laptop computer.

KEYWORDS

chemical shift prediction, NOESY, SIMPSON, simulated spectra, TOCSY

1 | INTRODUCTION

The nuclear spin interactions that we measure in nuclear magnetic resonance (NMR) experiments carry important information on the molecular structure and dynamics. This offers a very diverse range of applications of NMR. For example, NMR experiments are routinely used to elucidate the structure of molecules in organic synthesis, in more detailed studies of protein structures, as fingerprinting in metabolomics, in studies of polymers, or to characterize amorphous or crystalline materials. To obtain the desired information from NMR, the following approaches are the most prevalent. (a) Use of empirical knowledge based on experiments on similar compounds and (b) simulations based on Hamiltonian theory of the nuclear spin interactions.

The first approach is typically used for interpretation of NMR spectra of small organic molecules, where researchers in the early days established important empirical knowledge on ^1H and ^{13}C chemical shifts, for example, as gathered by Grant and Paul.^[1] Such studies have paved the way for more recent establishments of

databases for spectrum prediction.^[2–9] For proteins, similar approaches have been developed, starting from the pioneering work^[10] to programs predicting structural features of proteins from NMR spectra^[11] or providing random-coil shifts based on the protein sequence.^[12,13]

Simulations and theoretical descriptions of NMR pulse sequences by product operators^[14] have been essential for virtually all pulse-sequence developments. Although it is often possible to obtain insight in pulse sequences from analytical descriptions, it may be convenient to establish numerical simulations, for example, for large spin systems or powders with orientational effects. To achieve this, simulation programs such as SIMPSON,^[15–18] SpinEvolution,^[19] Spinach,^[20] and SpinDynamica^[21] have been developed. These versatile programs have been used to, for example, develop optimal control NMR experiments,^[22] investigation of proton-driven spin diffusion,^[23] and simulation of multidimensional NMR spectra of proteins.^[24]

In this tutorial manuscript, focus will be on coupling information obtained from different sources to provide new insight, as previously outlined for simulations of

so-called CHHC experiments^[25] for proteins in the solid state.^[26] Here, it will be demonstrated that simulated 1D and 2D total correlation spectroscopy (TOCSY) and nuclear Overhauser effect spectroscopy (NOESY) spectra of small proteins can easily be obtained using H and HA random-coil chemical shifts from a chemical-shift predictor^[13] or from the BioMagResBank,^[27] TOCSY transfer efficiencies calculated using SIMPSON,^[15–18] and NOESY cross-peak intensities obtained from average interatomic distances reported for ensembles of intrinsically disordered proteins^[28] analysed using visual molecular dynamics (VMD)^[29] and that these bits can be put together in a simple manner to obtain simulated multidimensional NMR spectra of peptides and proteins.

It should be emphasized that the goal of this work is not to provide the most accurate reproduction of experimental spectra or to establish new methods for spectrum prediction. The goal is to combine various computational and empirical methods to achieve, in a simple manner, simulated spectra that would otherwise require tremendous efforts to obtain. We use this approach to generate liquid-state NMR spectra of small peptides that students can use to practice the procedures for protein NMR spectrum assignment.

2 | SIMULATION METHODS

This section establishes the procedures for obtaining chemical shifts and J -couplings defining the nuclear spin systems as well as procedures for estimation of TOCSY and NOESY cross-peak intensities. The aim is that only the amino acid sequence should be added as input, then the scripts will return the simulated NMR spectra. As the focus is on small peptides, we assume that they adopt random-coil structures. However, the scripts are designed in a way that they can simulate the NMR spectra based on user-input chemical shift parameters.

We will use the tool command language (Tcl) scripting interface of SIMPSON^[15–18] to gather all information from different sources and generate the simulated spectra.

2.1 | Chemical shifts and J -couplings

The random-coil chemical shift predictor by Kjaergaard *et al.*^[13] has been selected, as it has a JavaScript interface that has been easy to adopt to the current Tcl setup. We have done this by creating a Tcl package, which allows the chemical shift predictor to be embedded in any Tcl script. The package is available for download from our web site.^[30] In Tcl (and thereby in a SIMPSON input file), a package is included by the following commands.

```
lappend ::auto_path /path/to/package
package require randomcoil
```

In random-coil structures, the $^3J_{\text{HNHA}}$ vary among the amino acids, as reported by Dobson and coworkers.^[31] We have included these J -coupling values in the random-coil package, so chemical shifts and J -couplings can be obtained for a specific sequence by the following commands.

```
set sequence "NFGAIL"
set shifts [randomcoil::shift $sequence 25
6.5]
set jcoupl [randomcoil::jhnha $sequence]
```

where the two additional inputs to the shift are temperature (in °C) and pH. As the predictor does not distinguish between the two alpha protons of glycine (Gly), we manually add and subtract 0.03 ppm to the predicted HA (HA refers to H α , HB to H β , etc. using conventional amino acid naming) chemical shift of Gly HA2 and HA3, respectively, as we know that there is a typical separation of 0.06 ppm between the two hydrogens.

The chemical shifts in the side chains are obtained from the BioMagResBank,^[27] which reports statistics on all chemical shifts, so we can obtain the average chemical shifts and their standard deviations.^[32] To establish the spin systems, three-bond J -couplings of $^3J_{\text{HH}} = 7$ Hz are added between all relevant hydrogens, and in cases where geminal hydrogens are different (neighbor to chiral carbon, i.e., in general HB2 and HB3, Ile HG12 and HG13, and Gly HA2 and HA3), $^2J_{\text{HH}} = 15$ Hz are added as well. The SIMPSON description of the spin systems for all 20 amino acids is given in the Supporting information. For fast generation of spin systems, we represent the spin system of alanine by the following definitions.

```
set atoms(ala) {{H 1} {HA 1} {HB 3}}
set coupling(ala) {{H HA 5.8} {HA HB 7}}
set shift(ala,H) {8.19 0.58}
set shift(ala,HA) {4.24 0.43}
set shift(ala,HB) {1.36 0.25}
```

The first line defines the three different atoms. For each atom is given the name and number of atoms, for alanine, there are three HB atoms. The second line lists the J -couplings in the amino acid, and the three last lines list the chemical shifts obtained from the BioMagResBank along with the standard deviations reported in the database. In SIMPSON format, the spin system will be defined as

```
spinsys {
  channels 1H
```

```
nuclei 1H 1H 1H 1H 1H
shift 1 8.19p 0 0 0 0 0
shift 2 4.24p 0 0 0 0 0
shift 3 1.36p 0 0 0 0 0
shift 4 1.36p 0 0 0 0 0
shift 5 1.36p 0 0 0 0 0
jcoupling 1 2 5.8 0 0 0 0 0
jcoupling 2 3 7.0 0 0 0 0 0
jcoupling 2 4 7.0 0 0 0 0 0
jcoupling 2 5 7.0 0 0 0 0 0
}
```

See Supporting information for a full list of spin systems for the amino acids.

2.2 | TOCSY cross-peak intensities

To establish the TOCSY cross-peak intensities, SIMPSON is to calculate the transfer efficiencies between the different hydrogen atoms in each amino acid using the spin systems as explained above. We have used the decoupling in the presence of scalar interactions (DIPSI)-2 pulse sequence,^[33] which in SIMPSON may be implemented as (remember that pulse lengths are in μ s).

```
proc dipsi2 {rf} {
  set tp [expr 0.25e6/$rf]

  foreach ph1 {x -x -x x} ph2 {-x x x -x} {
    pulse [expr $tp*3.556] $rf $ph1
    pulse [expr $tp*4.556] $rf $ph2
    pulse [expr $tp*3.222] $rf $ph1
    pulse [expr $tp*3.167] $rf $ph2
    pulse [expr $tp*0.333] $rf $ph1
    pulse [expr $tp*2.722] $rf $ph2
    pulse [expr $tp*4.167] $rf $ph1
    pulse [expr $tp*2.944] $rf $ph2
    pulse [expr $tp*4.111] $rf $ph1
  }
}
```

The total length of this block is 115.112 times the 90° pulse length. Hence, to achieve a certain mixing time (τ_{aum} , here 80 ms), we make the following use of the SIMPSON propagator command to repeat the DIPSI-2 block.

```
proc pulseseq {} {
  ...
  set taum 80000
  set ndipsi [expr int($taum/
(115.112*(0.25e6/$rf)))]
  dipsi2
```

```
store 1
prop 1 [expr $ndipsi-1]
store 1
...
}
```

which provides a simple and efficient way to calculate the TOCSY pulse sequence, as the total mixing sequence is now stored as Propagator 1 in SIMPSON.

The DIPSI-2 sequence transfers longitudinal magnetization. The goal is to figure out how much magnetization is transferred between the different hydrogen atoms in each amino acid. Therefore, we use SIMPSON to probe how this pulse sequence can transfer magnetization from one hydrogen to another, one by one. The most efficient way to implement this is by using the possibility to define the start and detect operators from within the pulse sequence.

```
proc pulseseq {} {
  ...
  for {set i 1} {$i <= $n} {incr i} {
    matrix set start operator I${i}z
    for {set j 1} {$j <= $i} {incr j} {
      matrix set detect operator I${j}z
      reset
      prop 1
      acq
    }
  }
}
```

This pulse sequence runs through a double loop where it sets the start operator to I_z on nucleus 1, 2, ..., n , assuming that the spin system has n hydrogens. The detect operator is likewise looped, and the transfer amplitude is sampled. The transfer amplitudes will later be read out in the main section of the SIMPSON input file, for example, by the code snippet.

```
proc main {} {
  ...
  set f [fsimpson]
  set ii 1
  for {set i 1} {$i <= $n} {incr i} {
    for {set j 1} {$j <= $i} {incr j} {
      puts "$i → $j: [findindex $f $ii -re]"
      incr ii
    }
  }
}
```

Table 1 shows the result of the simulation for Ile (11 hydrogen spins). Although this is a large spin system,

TABLE 1 Total correlation spectroscopy transfer efficiency between hydrogens in isoleucine for an 80 ms decoupling in the presence of scalar interactions-2 mixing block at 600 MHz using a radio frequency field strength of 10 kHz and radio frequency carrier frequency of 4 ppm

	H	HA	HB	HG12	HG13	HG21	HG22	HG23	HD11	HD12	HD13
H	11										
HA	53	13									
HB	10	6	19								
HG12	2	3	10	15							
HG13	2	3	10	23	15						
HG21	6	6	10	5	5	32					
HG22	6	6	10	5	5	13	32				
HG23	6	6	10	5	5	13	13	32			
HD11	1	2	5	10	10	3	3	3	32		
HD12	1	2	5	10	10	3	3	3	15	32	
HD13	1	2	5	10	10	3	3	3	15	15	32

the simulation may be performed in a few minutes using a normal desktop computer. Transfer efficiencies for all amino acids are given in the Supporting information.

2.3 | NOESY cross-peak intensities

Ignoring longitudinal relaxation, NOESY cross peaks appear due to cross-relaxation coupling nuclei I and S through this equation:^[34]

$$\frac{dI_z}{dt} = \sigma_{IS}(S_z^0 - S_z), \quad (1)$$

where σ_{IS} is the cross-relaxation rate that depends on the spectra density function and is proportional to the squared dipolar coupling. It may be hard to precisely quantify the NOESY cross-peak intensities due to the many parameters influencing the NOESY spectra, so here we assume that the cross-peak intensity can be described by

$$I_{IS} = 1 - \exp\left(-r_0^6 \left\langle \frac{1}{r_{IS}^6} \right\rangle \ln 2\right) \quad (2)$$

and to model a NOESY spectrum with a relatively short mixing time favouring short distances, we here choose a value of $r_0 = 3 \text{ \AA}$, which yields a cross-peak intensity of 50% for $r_{IS} = 3 \text{ \AA}$ and less than 1% at $r_{IS} = 6 \text{ \AA}$.

To establish average distances ($\langle r_{IS}^{-6} \rangle$), ensembles of structures for intrinsically disordered proteins have been used.^[28] As an example, Table 2 lists the average distances and transfer efficiencies based on Equation 2 for isoleucine, and Table 3 lists the interresidue distances required to do the sequential assignment. In the simulations, only the interresidue distances in the first column are included, as they are by far the most intense, in line with normal expectations from protein NMR.^[35]

2.4 | Simulation details

To simulate the 1D and 2D spectra of peptides and proteins, the calculation speed needs be considered. It is not convenient to perform full density-matrix calculations on large spin systems, although such efforts have been

TABLE 2 Average distances ($\langle r_{IS}^{-6} \rangle^{-1/6}$) and resulting transfer efficiencies calculated by Equation 2 with $r_0 = 3 \text{ \AA}$ for hydrogen atoms in isoleucine based on 30 ensemble structures of intrinsically disordered proteins (see text)^a

	H	HA	HB	HG12	HG13	HG2	HD1
H	100 ^b						
HA	2.94 / 54	100 ^b					
HB	3.07 / 45	2.50 / 87	100 ^b				
HG12	2.74 / 70	3.10 / 44	2.60 / 80	100 ^b			
HG13	3.09 / 44	3.35 / 30	2.44 / 91	1.77 / 100	100 ^b		
HG2	3.30 / 33	2.73 / 71	2.57 / 83	3.16 / 40	3.04 / 47	100 ^b	
HD1	2.13 / 100	3.90 / 13	3.15 / 40	2.59 / 81	2.57 / 83	2.92 / 56	100 ^b

^aThe data are represented as distance / transfer (in percent).

^bThe nuclear Overhauser effect spectroscopy diagonal peak intensities are set to 100%.

TABLE 3 Average distances ($\langle r_{IS}^{-6} \rangle^{-1/6}$) and resulting transfer efficiencies calculated by Equation 2 with $r_0 = 3 \text{ \AA}$ for interresidue contacts between hydrogen atoms in 30 ensemble structures of intrinsically disordered proteins (see text)^a

	H_{i+1}	HA_{i+1}	HB_{i+1}
H _i	2.89 / 58	4.96 / 3	5.36 / 2
HA _i	2.55 / 84	4.51 / 6	5.14 / 3
HB _i	3.13 / 42	4.68 / 5	5.37 / 2

^aThe data are represented as distance / transfer (in percent).

reported in a single case,^[24] so our approach relies only on such calculations to obtain the TOCSY transfer amplitudes. The actual simulation of the spectra is done simply by adding peaks at the positions of the chemical shifts of the different atoms. The coupling pattern is based on a simple splitting of the lines. As many couplings involve nuclei with similar chemical shifts (e.g., all the vicinal couplings), all couplings are handled as strong couplings, that is, a coupling between nuclei *a* and *b* gives rise to four peaks

$$\nu = \frac{\nu_a + \nu_b}{2} \pm \frac{Q}{2} \pm \frac{J}{2}, \quad (3)$$

where ν_a and ν_b are the chemical shifts, *J* is the *J*-coupling, and

$$Q = \sqrt{(\nu_a - \nu_b)^2 + J^2}. \quad (4)$$

The strongly coupled spin system approaches a normal 1:1 doublet, when the chemical shift difference is large (i.e., $Q \rightarrow |\nu_a - \nu_b|$) and shows the well-known roofing when the chemical shift difference is small (i.e., $Q \rightarrow J$, assuming a positive *J*-coupling). The intensity of the inner and outer peaks is given by

$$\text{Inner} = \frac{Q + J}{2Q}, \quad \text{Outer} = \frac{Q - J}{2Q}. \quad (5)$$

The coupling pattern for each nucleus is handled by nesting all its *J*-couplings through the normal *J*-coupling tree, where the second coupling splits the lines of the first coupling according to the peak positions and intensities given in Equations 3 and 5.

Taking HA2 of Gly3 in the peptide Pro–Ala–Gly–Tyr–Asn as an example, the HA2 chemical shift is assumed to be 3.95 ppm. We assume it has a $^3J_{\text{HNHA}}$ coupling of 5.8 Hz and a $^2J_{\text{HAHA}}$ coupling of 15 Hz. At 600 MHz, the cascade of couplings will be as follows, here listed as frequency (intensity)

$$2370.0(1.00) \rightarrow \begin{cases} 2367.1(0.50) \\ 2372.9(0.50) \end{cases} \rightarrow \begin{cases} 2361.1(0.35) \\ 2366.9(0.35) \\ 2376.1(0.15) \\ 2381.9(0.15) \end{cases} \quad (6)$$

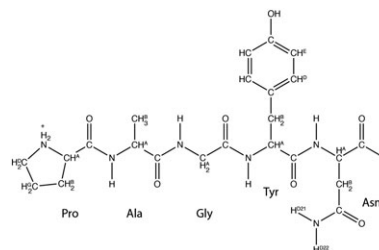


FIGURE 1 Molecular structure of the pentapeptide Pro–Ala–Gly–Tyr–Asn with indication of the naming convention for the different hydrogen atoms. Pro = proline; Ala = alanine; Gly = glycine; Tyr = tyrosine; Asn = asparagine

TABLE 4 Predicted ^1H chemical shifts for the peptide Pro–Ala–Gly–Tyr–Asn

Residue	Atom	Shift (ppm)
1 Pro	HA	4.45
	HB2	2.07
	HB3	2.00
	HG	1.92
	HD	3.65
2 Ala	H	8.49
	HA	4.25
	HB	1.36
3 Gly	H	8.33
	HA2	3.95
	HA3	3.89
4 Tyr	H	7.99
	HA	4.63
	HB2	2.90
	HB3	2.84
	HD	6.93
5 Asn	HE	6.70
	H	8.23
	HA	4.61
	HB2	2.80
	HB3	2.75
	HD21	7.32
	HD22	7.15

Note. Pro = proline; Ala = alanine; Gly = glycine; Tyr = tyrosine; Asn = asparagine.

and similarly for HA3. Each of these signals gives a doublet with significant roofing because of the strong coupling with HA3, which has a chemical shift of 3.89 ppm.

3 | RESULTS AND DISCUSSION

Focusing on small peptides, we will assume they adopt random-coil structures, but it would be straight forward to the present procedures with chemical shift predictors such as SHIFTX,^[36,37] SHIFTY,^[38] or PROSHIFT,^[39] if equipped with a known protein structure. In the current simulation setup, the flow is as follows:

1. Establish the spin system including chemical shifts from random-coil predictions or by user-input and J -couplings assuming standard values.
2. Simulate the coupling patterns in 1D spectrum.
3. Get TOCSY and NOESY transfer amplitudes by looking up the tables given in Supporting information.
4. Simulate the 2D spectra by extending the couplings in two dimensions

The SIMPSON script that performs these tasks is available for download from our web site.^[40] Alternatively, all files (the script and two Tcl libraries) are made available as Supporting information to this paper.

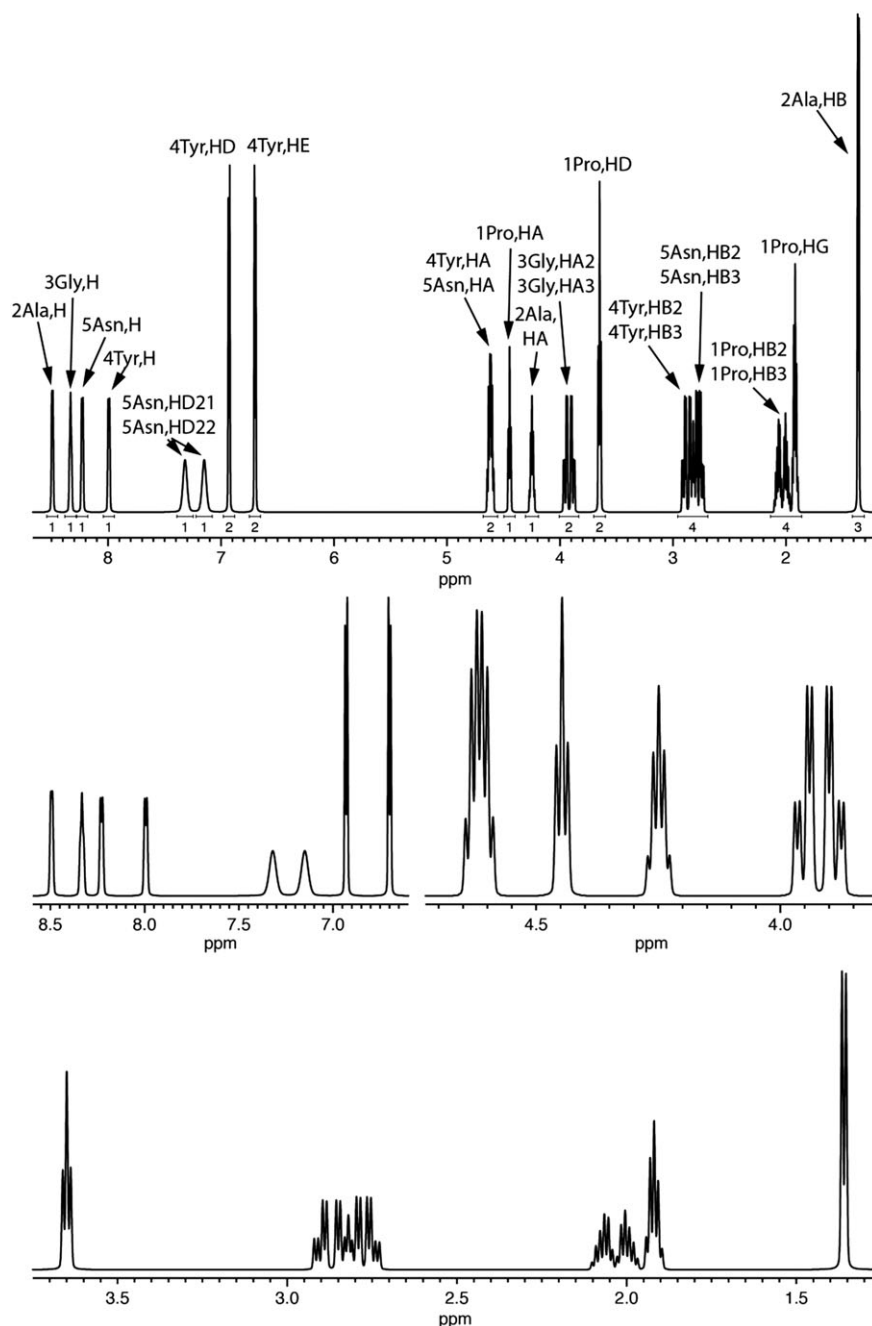


FIGURE 2 Simulated 1D spectrum of the peptide Pro—Ala—Gly—Tyr—Asn. The top row shows the full spectrum with indication of integrals below the spectrum. The two bottom rows are expansions of the amide-, alpha-, and aliphatic regions of the spectrum. Pro = proline; Ala = alanine; Gly = glycine; Tyr = tyrosine; Asn = asparagine

To illustrate the possibility to combine all this information, we will simulate ^1H 1D and 2D spectra of a pentapeptide with sequence Pro–Ala–Gly–Tyr–Asn. The molecular structure of this pentapeptide is shown in Figure 1, and the predicted chemical shifts are summarized in Table 4. The simulated 1D spectrum is shown in Figure 2 along with the assignment of each peak. By inspection of the coupling patterns, each of them indeed shows the expected couplings and multiplicities. Only the peaks of 4Tyr HA and 5Asn HA are so overlapping that it is not possible to distinguish the individual coupling patterns.

The 2D TOCSY and NOESY spectra shown in Figure 3 and with relevant expansions in Figure 4 clearly demonstrate the expected intraresidue cross peaks, and in the case of NOESY, also inter-residue cross peaks according to the short distances listed in Table 3. We often use this kind of spectra when teaching undergraduate students to assign NMR spectra of proteins, as they provide simple examples for the basic principles of both peak assignment and sequential assignment.

To compare the simulations with experimental spectra, Figure 5a shows the experimental ^1H NMR spectrum of the peptide NFGAIL, synthesized by solid-phase

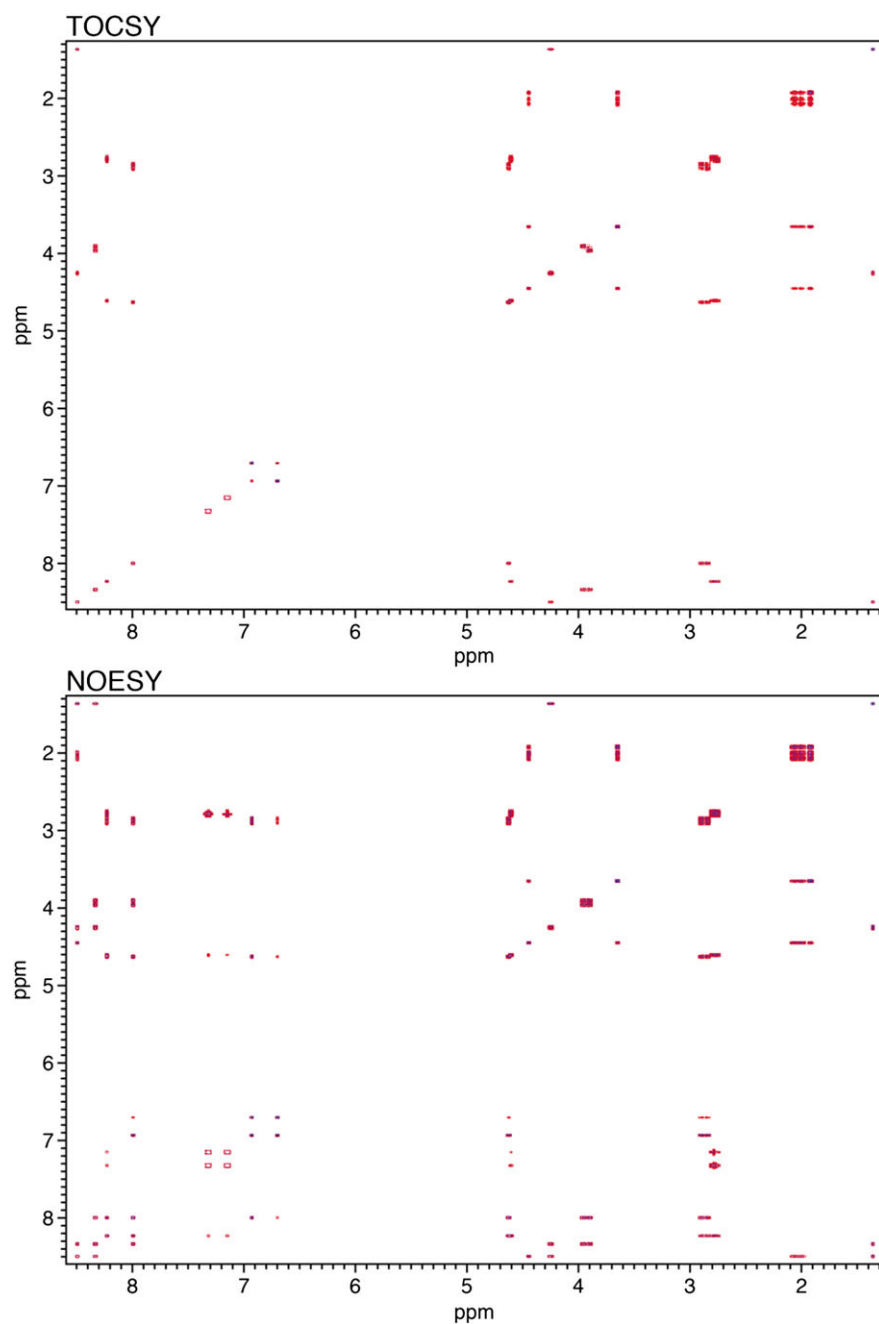


FIGURE 3 Simulated 2D total correlation spectroscopy (TOCSY) and nuclear Overhauser effect spectroscopy (NOESY) spectra of the peptide Pro–Ala–Gly–Tyr–Asn

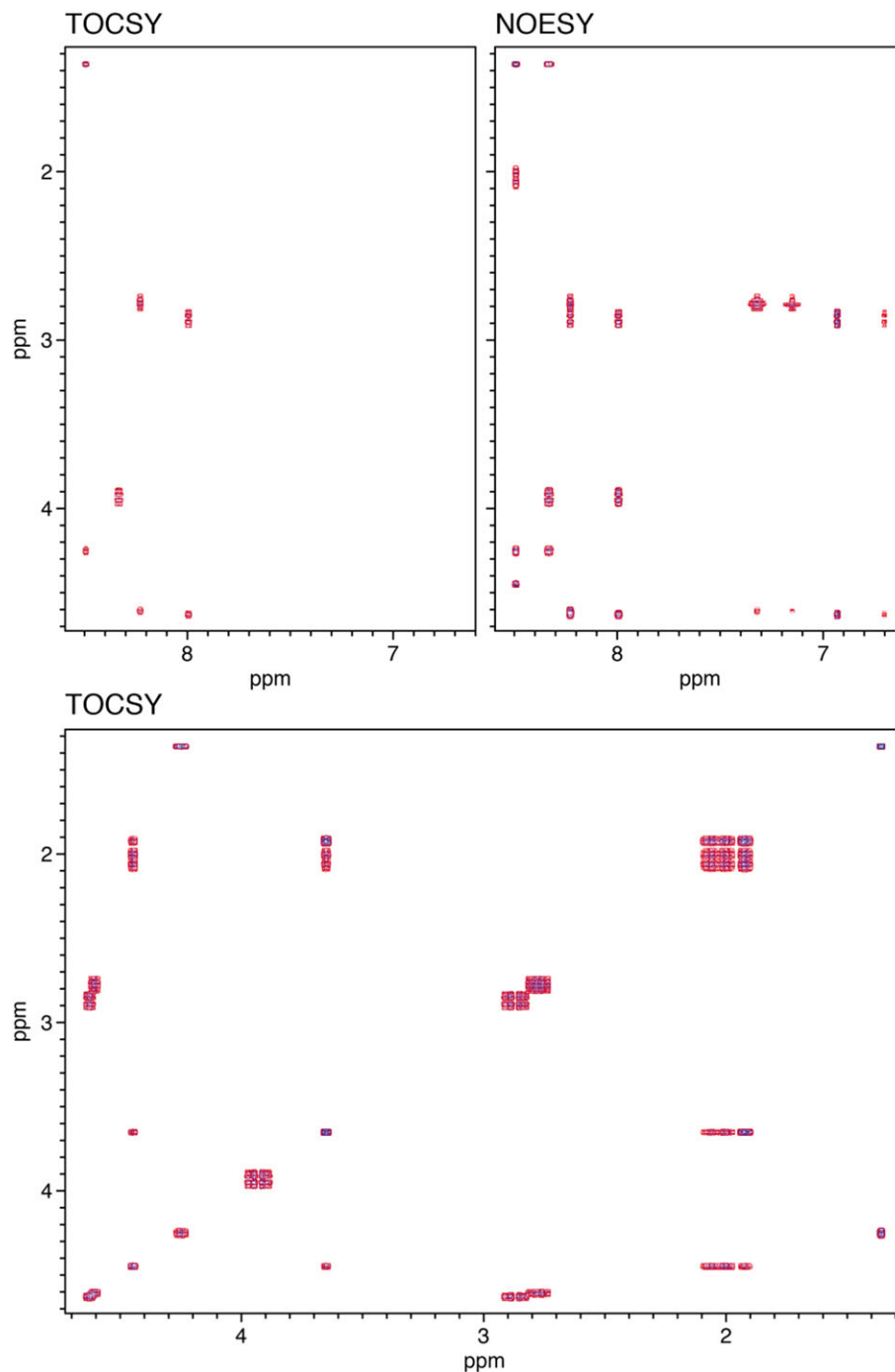


FIGURE 4 Expansions of the HN-HA fingerprint and aliphatic regions of the simulated total correlation spectroscopy (TOCSY) and nuclear Overhauser effect spectroscopy (NOESY) spectra of the peptide Pro-Ala-Gly-Tyr-Asn (top) and of the aliphatic TOCSY region (bottom)

peptide synthesis as described elsewhere.^[42,43] The spectrum was recorded on a Bruker spectrometer at 400.13 MHz in 9:1 H₂O:D₂O and using a pulse sequence with presaturation for suppression of the strong water signal. The assigned chemical shifts are listed in Table 5. Using these chemical shifts as input, our present simulation setup provided the simulated spectrum in Figure 5b, which very much resembles the experimental

spectrum. The major differences are varying intensities observed in the experimental spectrum, for example, from water suppression. These are not modelled in our setup, so the intensities differ from the experimental to the simulated spectrum for various signals.

As a final example of the present simulation setup, Figure 5c shows the simulated 2D TOCSY spectrum of ubiquitin employing the chemical shifts

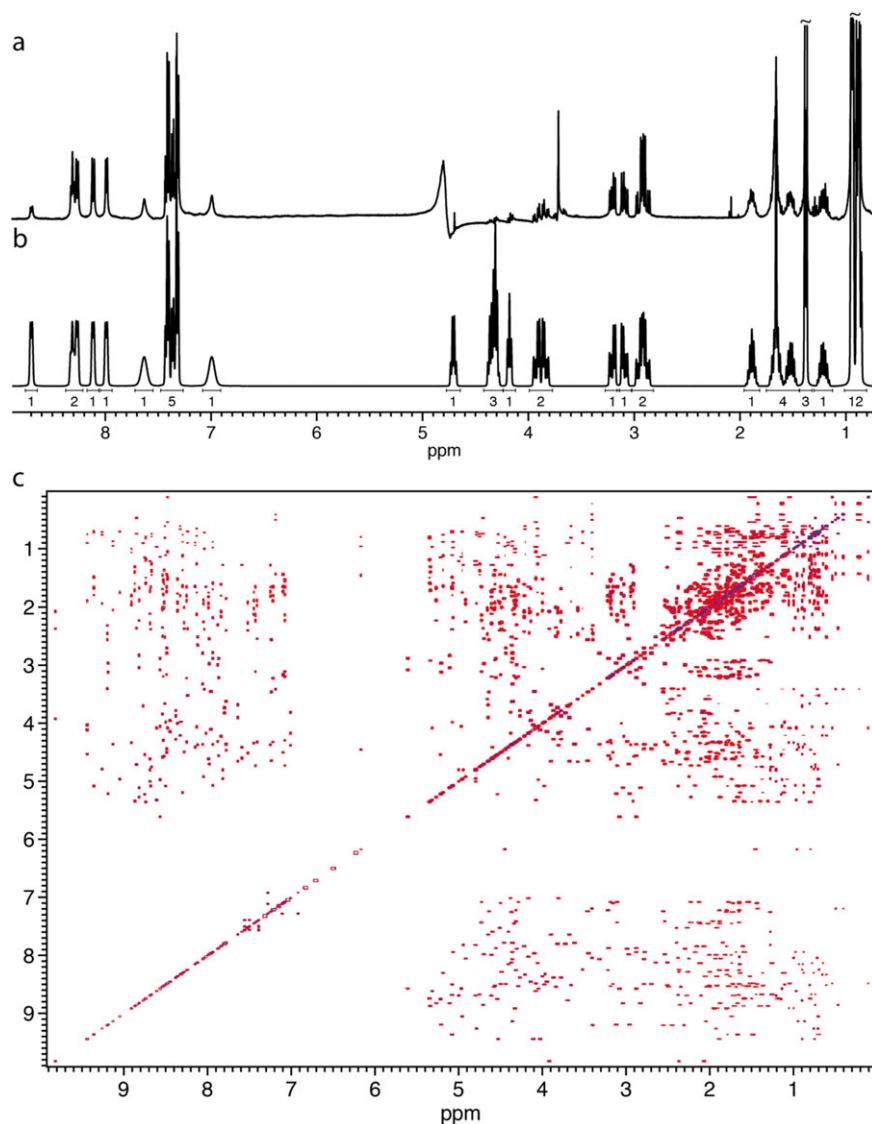


FIGURE 5 (a–b) Comparison between the experimental ^1H nuclear magnetic resonance spectrum (400.13 MHz) of the peptide Asn–Phe–Gly–Ala–Ile–Leu (a) and the corresponding simulated spectrum (b). (c) Simulated 600 MHz nuclear Overhauser effect spectroscopy spectrum of human ubiquitin based on assignments from an early work^[41]

reported in the BioMagResBank (BMRB id 68) from an early study.^[41]

Having demonstrated the possibility to simulate spectra of peptides and small proteins, it is relevant to evaluate the time it takes to perform such simulations. The computation time mainly consists of two contributions: (a) calculation of the peaks in the 2D spectra and (b) writing the 2D datasets to the disk. The latter part takes a few seconds, and the first part depends on the number of amino acids in the sequence. In fact, the length of this part is proportional to the number of amino acids, as there are no long-range effects included in the simulations. Simple tests show that simulations of up to proteins with up to 100 amino acids can be done in less than a minute of computation time on a normal laptop computer, whereas short peptides of less than 10 amino acids may be simulated in a few seconds.

The fact that the simulations are very fast is encouraging for making more precise estimates of transfer amplitudes in the multidimensional spectra and intensities in the 1D spectrum, because actual simulations of full multidimensional spectra as part of, for example, fitting a set of structural parameters is indeed within reach using the present concept of separating the computation-dense efforts from the full spectrum simulation.

Next project in line is to merge the present spectrum simulation into our new web-based spectrum analysis interface, for which we have already launched the spectrum viewer jsNMR^[44] and a spectroscopy game. A test site for the web-based version of the program may be found on our web site.^[45] Once the implementation is complete, it will be available together with all scripts employed in this work.

TABLE 5 Experimentally determined ^1H chemical shifts for the peptide Asn–Phe–Gly–Ala–Ile–Leu

Residue	Atom	Shift (ppm)
1 Asn	HA	4.314
	HB2	2.954
	HB3	2.885
	HD21	7.633
	HD22	6.994
2 Phe	H	8.699
	HA	4.710
	HB2	3.207
	HB3	3.093
	HD	7.318
	HE	7.416
3 Gly	HZ	7.355
	H	8.312
	HA2	3.924
4 Ala	HA3	3.840
	H	7.991
	HA	4.314
5 Ile	HB	1.380
	H	8.116
	HA	4.180
	HB	1.889
	HG12	1.526
	HG13	1.213
6 Leu	HG2	0.937
	HD1	0.871
	H	8.267
	HA	4.358
	HB2	1.680
	HB3	1.640
	HG	1.660
	HD1	0.944
	HD2	0.891

Note. Asn = asparagine; Phe = phenylalanine; Gly = glycine; Ala = alanine; Ile = isoleucine; Leu = leucine.

4 | CONCLUSIONS

This paper demonstrates how information obtained from different sources (chemical shift prediction or BioMagResBank, density-matrix simulations, and average distances from PDB structures) may easily be put together to obtain simulated one-dimensional and multidimensional NMR spectra of peptides and proteins in solution.

The outlined methods provide quite accurate simulations that may easily be improved by adding better chemical shift prediction and/or more empirical information, for example, about typical TOCSY and NOESY transfer intensities.

ACKNOWLEDGEMENTS

Financial support from the Danish Ministry of Higher Education and Science (Grant AU-2010-612-181), Carlsbergfondet, the European Commission (H2020 Future and Emerging Technologies, Grant 731475) and (H2020 Research Infrastructures, Grant 731005), and Aarhus University is acknowledged. Thanks to Drs Jakob Toudahl Nielsen and Berit Paaske for useful discussions related to this project.

ORCID

Thomas Vosegaard  <http://orcid.org/0000-0001-5414-4550>

REFERENCES

- [1] D. M. Grant, E. G. Paul, *J. Am. Chem. Soc.* **1964**, 86, 2984.
- [2] J. Aires-de-Sousa, M. C. Hemmer, J. Gasteiger, *Anal. Chem.* **2002**, 74, 80.
- [3] Y. Binev, M. M. B. Marques, J. Aires-de-Sousa, *J. Chem. Inf. Model.* **2007**, 47, 2089.
- [4] A. M. Castillo, A. Bernal, R. Dieden, L. Patiny, J. Wist, *Aust. J. Chem.* **2016**, 8.
- [5] *ACD/HNMR P predictor*; advanced chemistry developments.
- [6] SPINUS-WEB: Prediction of NMR spectra - ^1H NMR chemical shifts, Coupling Constants and Full Spectrum from the Molecular Structure <http://www2.ccc.uni-erlangen.de/services/spinus/> (accessed Jun 14, 2017).
- [7] Modgraph NMR prediction http://www.modgraph.co.uk/product_nmr.htm (accessed Jun 14, 2017).
- [8] *ChemDraw v16.0*; PerkinElmer. <http://www.cambridgesoft.com/software/overview.aspx>.
- [9] Predict ^1H NMR spectra http://www.cheminfo.org/Spectra/NMR/Predictions/1H_Prediction/index.html (accessed Jun 14, 2017).
- [10] K. Wüthrich, *NMR of Proteins and Nucleic Acids*, John Wiley & Sons, New York **1986**.
- [11] Y. Shen, F. Delaglio, G. Cornilescu, A. Bax, *J. Biomol. NMR* **2009**, 44, 213.
- [12] K. Tamiola, B. Acar, F. A. A. Mulder, *J. Am. Chem. Soc.* **2010**, 132, 18000.
- [13] M. Kjaergaard, S. Brander, F. M. Poulsen, *J. Biomol. NMR* **2011**, 49, 139.
- [14] O. W. Sørensen, G. W. Eich, M. H. Levitt, G. Bodenhausen, R. R. Ernst, *Prog. Nucl. Magn. Reson. Spectrosc.* **1984**, 16, 163.

- [15] M. Bak, J. T. Rasmussen, N. C. Nielsen, *J. Magn. Reson.* **2000**, *147*, 296.
- [16] T. Vosegaard, A. Malmendal, N. C. Nielsen, *Monatsh. Chem.* **2002**, *133*, 1555.
- [17] Z. Tosner, T. Vosegaard, C. Kehlet, N. Khaneja, S. J. Glaser, N. C. Nielsen, *J. Magn. Reson.* **2009**, *197*, 120.
- [18] Z. Tošner, R. Andersen, B. Stevansson, M. Edén, N. C. Nielsen, T. Vosegaard, *J. Magn. Reson.* **2014**, *246*, 79.
- [19] M. Veshtort, R. G. Griffin, *J. Magn. Reson.* **2006**, *178*, 248.
- [20] H. J. Hogben, M. Krzystyniak, G. T. P. Charnock, P. J. Hore, I. Kuprov, *J. Magn. Reson.* **2011**, *208*, 179.
- [21] SpinDynamica » NMR Calculation and Simulation in Mathematica <http://www.spindynamica.soton.ac.uk/> (accessed Jun 13, 2017).
- [22] C. T. Kehlet, A. C. Sivertsen, M. Bjerring, T. O. Reiss, N. Khaneja, S. J. Glaser, N. C. Nielsen, *J. Am. Chem. Soc.* **2004**, *126*, 10202.
- [23] M. Veshtort, R. G. Griffin, *J. Chem. Phys.* **2011**, *135*, 134509.
- [24] D. V. Savostyanov, S. V. Dolgov, J. M. Werner, I. Kuprov, *Phys Rev B* **2014**, *90* 085139.
- [25] M. Aluas, C. Tripon, J. M. Griffin, X. Filip, V. Ladizhansky, R. G. Griffin, S. P. Brown, C. Filip, *J. Magn. Reson.* **2009**, *199*, 173.
- [26] T. Vosegaard, *Solid State Nucl. Magn. Reson.* **2010**, *38*, 77.
- [27] E. L. Ulrich, H. Akutsu, J. F. Doreleijers, Y. Harano, Y. E. Ioannidis, J. Lin, M. Livny, S. Mading, D. Maziuk, Z. Miller, E. Nakatani, C. F. Schulte, D. E. Tolmie, R. Kent Wenger, H. Yao, J. L. Markley, *Nucleic Acids Res.* **2007**, *36*, D402.
- [28] M. Varadi, S. Kosol, P. Lebrun, E. Valentini, M. Blackledge, A. K. Dunker, I. C. Felli, J. D. Forman-Kay, R. W. Kriwacki, R. Pierattelli, J. Sussman, D. I. Svergun, V. N. Uversky, M. Vendruscolo, D. Wishart, P. E. Wright, P. Tompa, *Nucleic Acids Res.* **2014**, *42*, D326.
- [29] W. Humphrey, A. Dalke, K. Schulten, *J Mol Graph* **1996**, *14*, 33.
- [30] Tcl libraries <http://nmr.au.dk/software/tcl-libraries/> (accessed Jul 24, 2017).
- [31] L. J. Smith, K. A. Bolin, H. Schwalbe, M. W. MacArthur, J. M. Thornton, C. M. Dobson, *J. Mol. Biol.* **1996**, *255*, 494.
- [32] Full chemical shift statistics http://www.bmrwisc.edu/ref_info/statsel.htm (accessed Jun 15, 2017).
- [33] J. Cavanagh, M. Rance, *J. Magn. Reson.* **1969** **1992**, *96*, 670.
- [34] J. Keeler, *Understanding NMR Spectroscopy*, 2nd ed., Wiley, Chichester **2010**.
- [35] J. Cavanagh, W. J. Fairbrother, A. G. Palmer III, N. J. Skelton, *Protein NMR Spectroscopy: Principles and Practice*, Academic Press, USA **1995**.
- [36] S. Neal, A. M. Nip, H. Zhang, D. S. Wishart, *J. Biomol. NMR* **2003**, *26*, 215.
- [37] B. Han, Y. Liu, S. W. Ginzing, D. S. Wishart, *J. Biomol. NMR* **2011**, *50*, 43.
- [38] D. S. Wishart, M. S. Watson, R. F. Boyko, B. D. Sykes, *J. Biomol. NMR* **1997**, *10*, 329.
- [39] J. Meiler, *J. Biomol. NMR* **2003**, *26*, 25.
- [40] Instant simulation of protein NMR spectra <http://nmr.au.dk/software/simpson-code-examples?Program=SIMPSON&mode=show&id=33> (accessed Jul 24, 2017).
- [41] P. L. Weber, S. C. Brown, L. Mueller, *Biochemistry (Moscow)* **1987**, *26*, 7282.
- [42] J. T. Nielsen, M. Bjerring, M. D. Jeppesen, R. O. Pedersen, J. M. Pedersen, K. L. Hein, T. Vosegaard, T. Skrydstrup, D. E. Otzen, N. C. Nielsen, *Angew Chem Int Ed* **2009**, *48*, 2118.
- [43] C. U. Hjørringgaard, J. M. Pedersen, T. Vosegaard, N. C. Nielsen, T. Skrydstrup, *J Org Chem* **2009**, *74*, 1329.
- [44] T. Vosegaard, *Magn. Reson. Chem.* **2015**, *53*, 285.
- [45] Instant simulation of 2D protein NMR spectra <http://jsnmr3.pastis.dk/protein.html> (accessed Jul 24, 2017).

SUPPORTING INFORMATION

Additional Supporting Information may be found online in the supporting information tab for this article.

How to cite this article: Vosegaard T. Fast simulations of multidimensional NMR spectra of proteins and peptides. *Magn Reson Chem.* 2018;56:438–448. <https://doi.org/10.1002/mrc.4663>